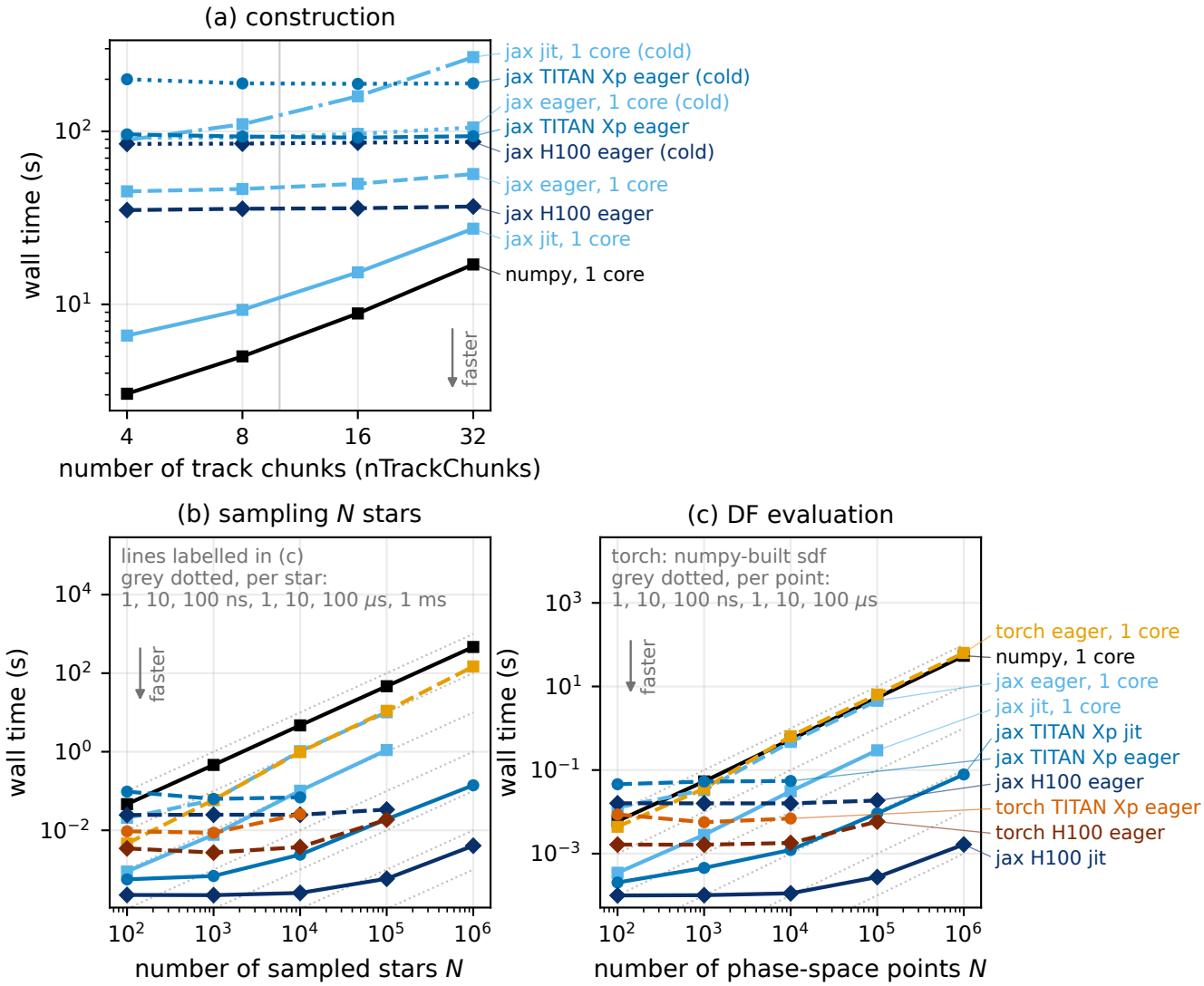
(a) construction
wall time (s)
$10^2$
$10^1$
jax jit, 1 core (cold)
jax TITAN Xp eager (cold)
jax eager, 1 core (cold)
jax TITAN Xp eager
jax H100 eager (cold)
jax eager, 1 core
jax H100 eager
jax jit, 1 core
numpy, 1 core
faster
4
8
16
32
number of track chunks (nTrackChunks)
(b) sampling $N$ stars
lines labelled in (c)
grey dotted, per star:
1, 10, 100 ns, 1, 10, 100 $\mu$s, 1 ms
wall time (s)
$10^4$
$10^2$
$10^0$
$10^{-2}$
faster
$10^2$
$10^3$
$10^4$
$10^5$
$10^6$
number of sampled stars $N$
(c) DF evaluation
torch: numpy-built sdf
grey dotted, per point:
1, 10, 100 ns, 1, 10, 100 $\mu$s
wall time (s)
$10^3$
$10^1$
$10^{-1}$
$10^{-3}$
faster
$10^2$
$10^3$
$10^4$
$10^5$
$10^6$
number of phase-space points $N$
torch eager, 1 core
numpy, 1 core
jax eager, 1 core
jax jit, 1 core
jax TITAN Xp jit
jax TITAN Xp eager
jax H100 eager
torch TITAN Xp eager
torch H100 eager
jax H100 jit